X-ray Fluorescence / Moseley's Law Analysis

Data analyzed:
- Reference spectrum: FeZn
- Unknown / sample spectra: W, Au, Pb, U

Workflow used:
1. Fit Fe and Zn Kα/Kβ peaks in the FeZn spectrum.
2. Calibrate the MCA with $E = a\,n + b$ from the four reference lines.
3. Fit the sample peaks with Gaussian models to extract channel centers.
4. Convert centers to energies and build Moseley plots using Eq. (7).

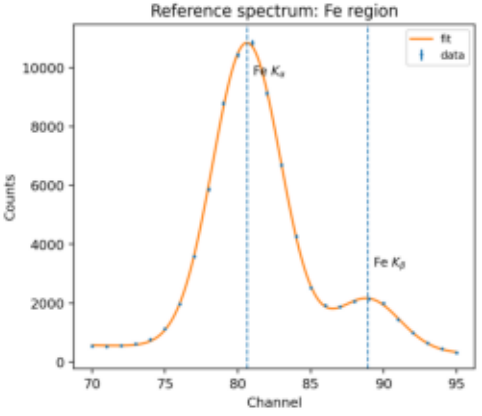
Calibration:

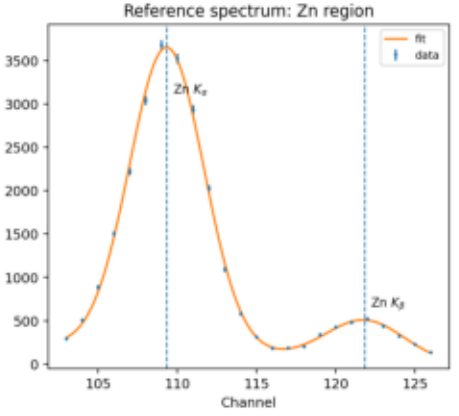
$$n = (12.8776 \pm 0.0108)\,E + (-1.798 \pm 0.076)$$

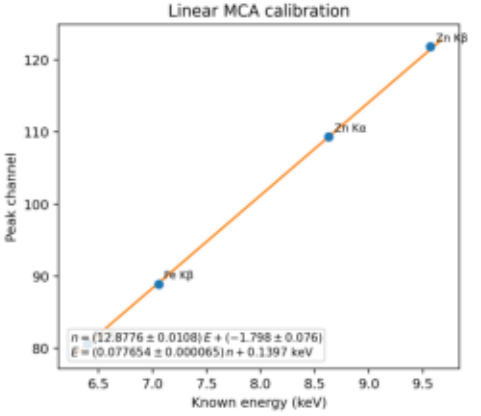
$$E = (0.077654)\,n + (0.1397)\ \text{keV}$$

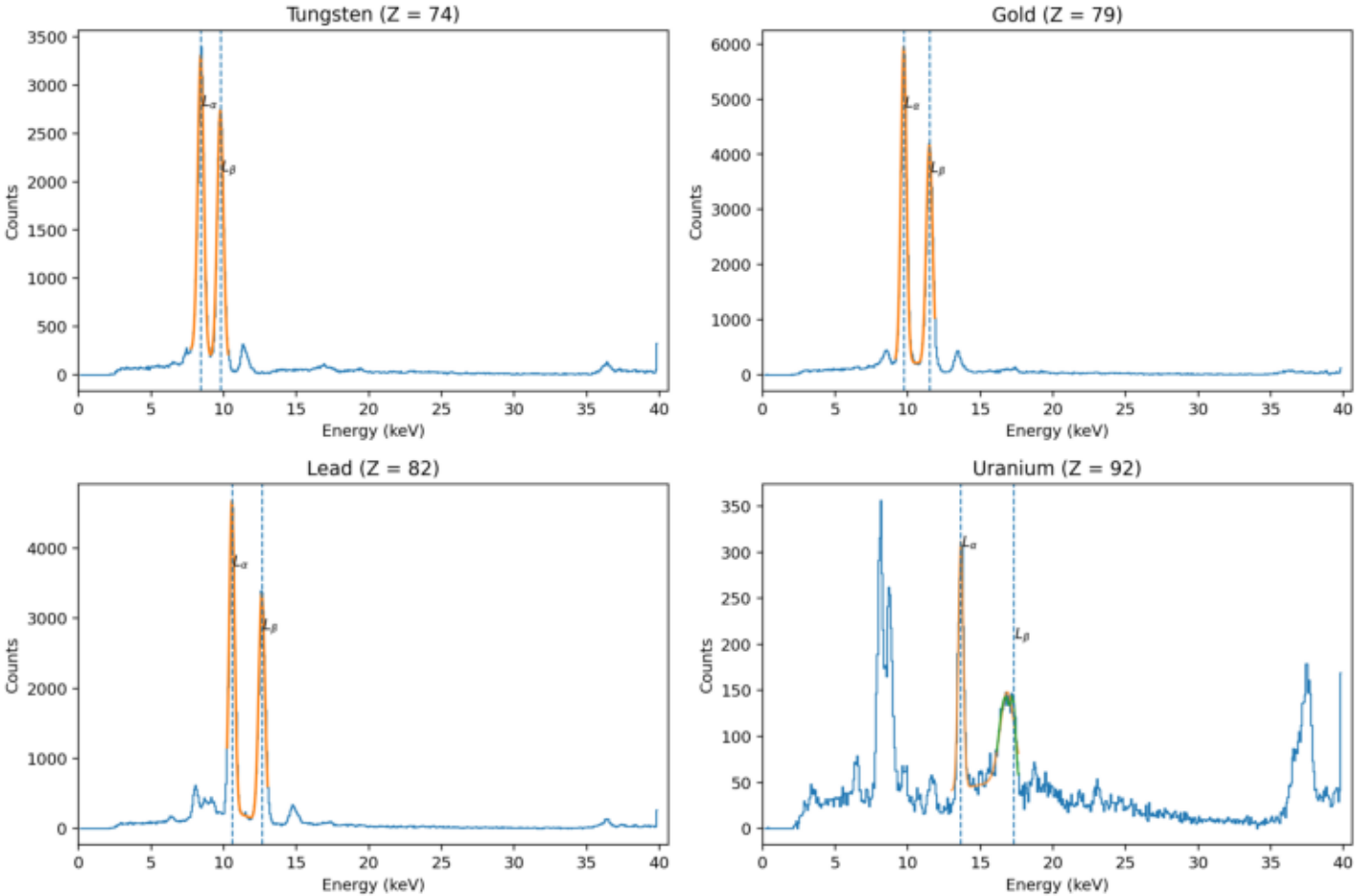
Screening constants from Eq. (7) (fixed unit-slope form):
- Kα: $\sigma = 0.944 \pm 0.018$
- Kβ: $\sigma = 1.855 \pm 0.020$
- Lα: $\sigma = 7.229 \pm 0.061$
- Lβ: $\sigma = 11.918 \pm 0.557$

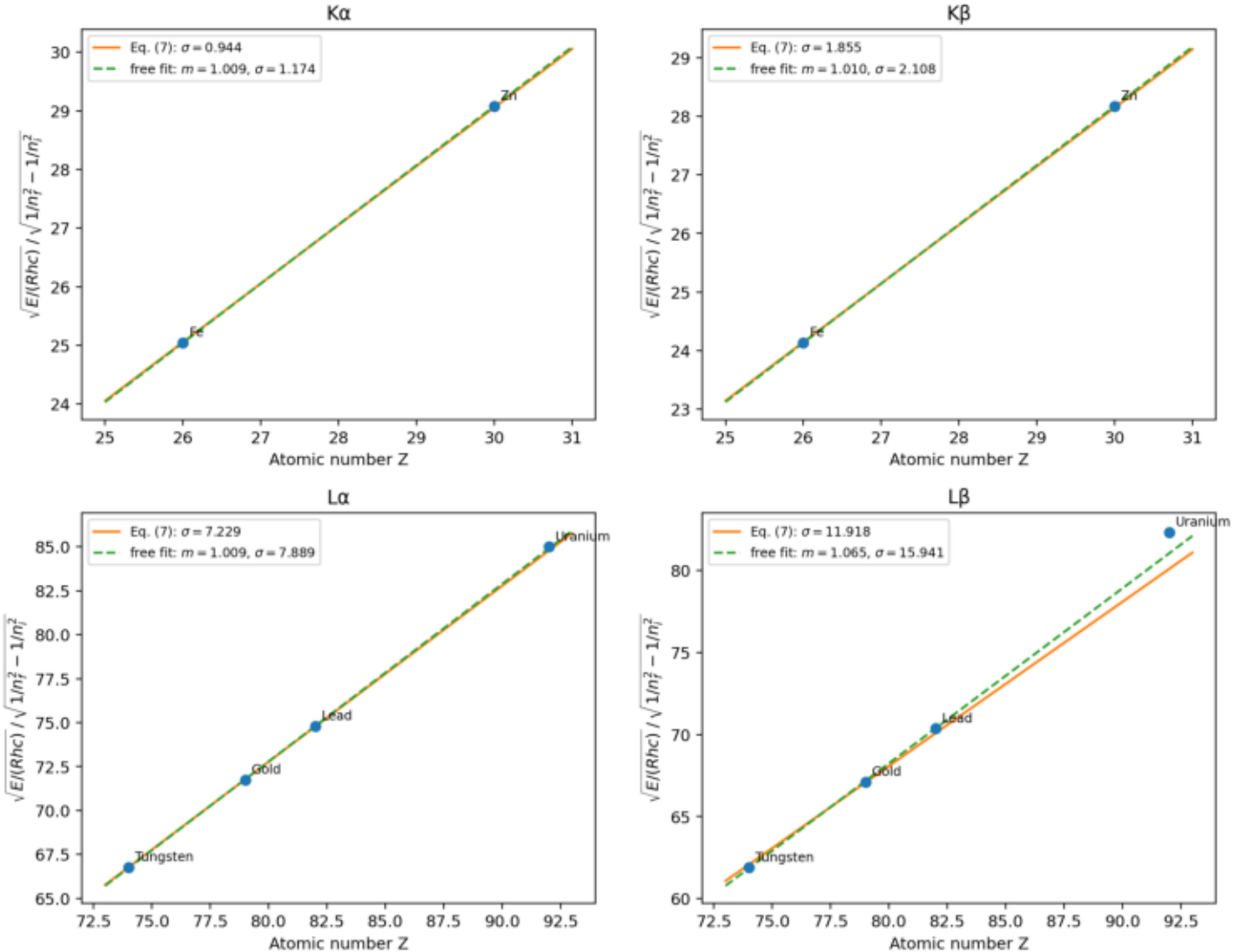
Notes:
- Only two K-series points are available (Fe and Zn), so the Kα/Kβ plots are underconstrained.
- The uranium Lβ region shows visible substructure; the higher-energy subpeak was used for the Lβ point.
- The quoted σ uncertainty is the across-point scatter standard error, which is more honest here than the tiny formal fit error.


Reference spectrum: Fe region
fit
data
Fe K_α
Fe K_β
Counts
Channel
0
2000
4000
6000
8000
10000
70
75
80
85
90
95


Reference spectrum: Zn region
fit
data
Zn K_α
Zn K_β
Channel
0
500
1000
1500
2000
2500
3000
3500
105
110
115
120
125


Linear MCA calibration
Zn Kβ
Zn Kα
Fe Kβ
$n = (12.8776 \pm 0.0108)E + (-1.798 \pm 0.076)$
$E = (0.077654 \pm 0.000065)n + 0.1397$ keV
Peak channel
Known energy (keV)
80
90
100
110
120
6.5
7.0
7.5
8.0
8.5
9.0
9.5


Tungsten (Z = 74)
Gold (Z = 79)
Lead (Z = 82)
Uranium (Z = 92)
Counts
Energy (keV)
L_α
L_β


Kα
Eq. (7): σ = 0.944
free fit: m = 1.009, σ = 1.174
Zn
Fe
Atomic number Z
Kβ
Eq. (7): σ = 1.855
free fit: m = 1.010, σ = 2.108
Zn
Fe
Atomic number Z
Lα
Eq. (7): σ = 7.229
free fit: m = 1.009, σ = 7.889
Uranium
Lead
Gold
Tungsten
Atomic number Z
Lβ
Eq. (7): σ = 11.918
free fit: m = 1.065, σ = 15.941
Uranium
Lead
Gold
Tungsten
Atomic number Z

Extracted line centers and energies

sample	Z	line	channel	channel_unc	energy_keV	energy_unc_keV
Fe	26	Kα	80.6259	0.0127	6.4006	0.0014
Zn	30	Kα	109.3265	0.0222	8.6293	0.0024
Fe	26	Kβ	88.9103	0.0447	7.0439	0.0036
Zn	30	Kβ	121.8252	0.0825	9.5999	0.0068
Tungsten	74	Lα	106.6568	0.0230	8.4220	0.0023
Gold	79	Lα	123.4254	0.0154	9.7242	0.0027
Lead	82	Lα	134.3272	0.0181	10.5707	0.0034
Uranium	92	Lα	174.0255	0.0774	13.6535	0.0082
Tungsten	74	Lβ	124.0550	0.0250	9.7730	0.0031
Gold	79	Lβ	146.1753	0.0204	11.4908	0.0042
Lead	82	Lβ	160.8363	0.0221	12.6293	0.0051
Uranium	92	Lβ	220.8234	0.4542	17.2875	0.0363